



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2021 Year 12 Course Assessment Task 4 (Trial Examination)

Thursday September 2, 2021

General instructions

- Working time – 2 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 12)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Mr Sekaran

☐ 12MAX.2 – Mrs Bhamra

☐ 12MXX.2 – Ms Ham

☐ 12MAX.3 – Mr Lam

☐ 12MAX.1 – Mr Ho

☐ 12MAX.4 – Mr Sun

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{16}$	$\overline{16}$	$\overline{14}$	$\overline{14}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

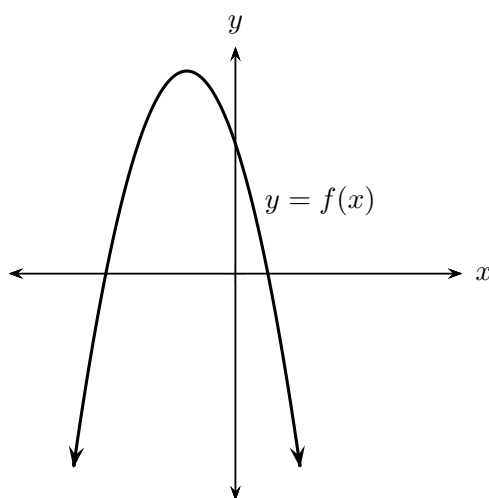
Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 12).

Questions

Marks

1. The graph of $y = f(x)$ is shown. Consider the largest possible domain for which $y = f(x)$ has an inverse function $y = f^{-1}(x)$. 1



How many points of intersection are there between $y = f(x)$ and $y = f^{-1}(x)$?

- (A) 1 (C) 3
 (B) 2 (D) 4
2. Which of the following has the same solutions as the inequality 1

$$\frac{-x}{(x-3)(2x+1)} \leq \frac{1}{4}$$

- (A) $(2x-3)(x+1)(x-3)(2x+1) \geq 0$ (C) $(2x-3)(x+1) \geq 0$
 (B) $(2x-3)(x+1)(x-3)(2x+1) \leq 0$ (D) $(2x-3)(x+1) \leq 0$

Examination continues overleaf...

3. The letters in the word CALCULUS were randomly rearranged. 1

What is the probability that C is at the start and end of the new arrangement?

- (A) $\frac{1}{6}$ (C) $\frac{1}{28}$
 (B) $\frac{1}{14}$ (D) $\frac{1}{56}$

4. Let $P(x) = x^3 + bx^2 + 6x + d$, where b and d are real numbers. 1

Given that $(x - 2)^2$ is a factor of $P(x)$, which of the following are possible values of b and d ?

- (A) $b = -6$, $d = 4$ (C) $b = \frac{9}{2}$, $d = 2$
 (B) $b = 6$, $d = -4$ (D) $b = -\frac{9}{2}$, $d = -2$

5. Which of the following is the derivative of $y = -\cos^{-1}(\ln x)$? 1

- (A) $\frac{-1}{x\sqrt{1-2\ln x}}$ (C) $\frac{1}{x\sqrt{1-2\ln x}}$
 (B) $\frac{-1}{x\sqrt{(1+\ln x)(1-\ln x)}}$ (D) $\frac{1}{x\sqrt{(1+\ln x)(1-\ln x)}}$

6. A curve is represented by the parametric equations $x = 5t^3 - 7t$ and $y = 2 + 3t - 4t^2$. 1

Which of the following is an expression for $\frac{dy}{dx}$ in terms of t ?

- (A) $3 - 8t$ (C) $(15t^2 - 7)(3 - 8t)$
 (B) $\frac{3 - 8t}{15t^2 - 7}$ (D) $\frac{15t^2 - 7}{3 - 8t}$

7. Which of the following is the unit vector perpendicular to $\underline{p} = -6\underline{i} + 2\underline{j}$? 1

- (A) $\underline{q} = \underline{i} + 3\underline{j}$ (C) $\underline{q} = \frac{1}{\sqrt{10}}\underline{i} + \frac{-3}{\sqrt{10}}\underline{j}$
 (B) $\underline{q} = \frac{1}{\sqrt{10}}\underline{i} + \frac{3}{\sqrt{10}}\underline{j}$ (D) $\underline{q} = \frac{-3}{\sqrt{10}}\underline{i} + \frac{1}{\sqrt{10}}\underline{j}$

Examination continues overleaf...

8. Which of the following is equivalent to $\int \sin 3x \sin 5x \, dx$? 1

(A) $\frac{1}{16} \sin 8x - \frac{1}{4} \sin 2x + c$ (C) $\frac{1}{4} \sin 2x - \frac{1}{16} \sin 8x + c$

(B) $\frac{1}{8} \sin 8x - \frac{1}{2} \sin 2x + c$ (D) $\frac{1}{2} \sin 2x - \frac{1}{8} \sin 8x + c$

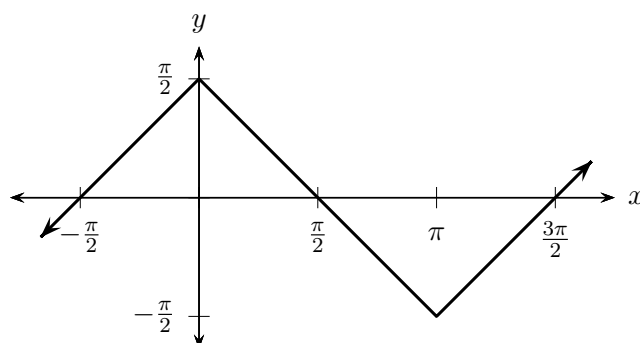
9. Which of the following integrals is equivalent to the integral I ? Use the substitution $u = 3 - 2x^2$. 1

$$I = \int_0^2 \frac{5x}{(3 - 2x^2)^4} \, dx$$

(A) $\frac{5}{4} \int_{-5}^3 \frac{1}{u^4} \, du$ (C) $\frac{5}{\sqrt{2}} \int_{-5}^3 \frac{\sqrt{3-u}}{u^4} \, du$

(B) $-\frac{5}{4} \int_{-5}^3 \frac{1}{u^4} \, du$ (D) $-\frac{5}{\sqrt{2}} \int_{-5}^3 \frac{\sqrt{3-u}}{u^4} \, du$

10. Which of the following equations is shown in the sketch below? 1



- (A) $y = \cos^{-1}(\sin x)$ (C) $y = \cos(\sin^{-1} x)$
 (B) $y = \sin^{-1}(\cos x)$ (D) $y = \sin^{-1}(\sin x)$

Examination continues overleaf...

Section II

60 marks

Attempt Question 11 to 14

Allow approximately 1 hour 45 minutes for this section

Question 11 (16 Marks)

Commence a NEW Writing Booklet

Marks

- (a) A class of fifteen mathematics students are seated around a circular table to discuss mathematical problems.
- In how many different ways can the students be arranged? **1**
 - If the seats are randomly assigned, find the probability that four particular students, Aidan, Christopher, Daniel and Matthew, are not all seated together as a group of four? Give your answer in the simplest form. **2**
- (b) The polynomial equation $P(x) = ax^3 + 12x^2 + cx + d$ has roots α , β and γ , where a , c and d are real numbers.
- Given that $\alpha + \beta + \gamma = 6$ and $\alpha^2 + \beta^2 + \gamma^2 = 32$, find the values of a and c . **2**
 - Given that the polynomial can be expressed as $P(x) = (x + 2)Q(x)$ where $Q(x)$ is a polynomial, find the value of d . **1**
- (c) Using the substitution $t = \tan \theta$, solve for $0^\circ \leq \theta \leq 360^\circ$ **3**

$$2 \sin 2\theta = 2 + 4 \cos 2\theta$$

- (d) The coefficients of the x^{17} and the x^{27} terms are equal in the expansion of **3**

$$\left(px^3 - \frac{q}{x^2} \right)^9$$

Prove that $p - 6q = 0$, where p and q are positive integers.

- (e) i. Show that $\frac{1}{(n+2)!} - \frac{n+1}{(n+3)!} = \frac{2}{(n+3)!}$, where n is an integer. **1**
- ii. Prove by mathematical induction that, for all positive integers n , **3**

$$\frac{1 \times 2}{3!} + \frac{2 \times 2^2}{4!} + \frac{3 \times 2^3}{5!} + \cdots + \frac{n \times 2^n}{(n+2)!} = 1 - \frac{2^{n+1}}{(n+2)!}$$

Examination continues overleaf...

Question 12 (16 Marks)

Commence a NEW Writing Booklet

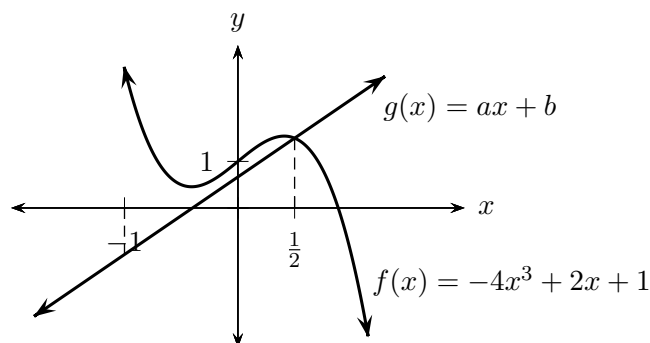
Marks

(a) Consider $f(x) = 4 \cos^{-1} \left(\frac{1}{6}x \right) - \frac{\pi}{2}$.

i. State the domain and range of $f(x)$. **2**

ii. Hence sketch the graph of $f(x)$. (Do NOT find the x -intercepts) **2**

(b) The graphs of $f(x) = -4x^3 + 2x + 1$ and $g(x) = ax + b$ are shown.



i. Sketch the graph for $h(x) = -4|x|^3 + 2|x| + 1$. **1**

ii. The solution to the inequality $ax + b > -4|x|^3 + 2|x| + 1$ is **2**

$$x \in (-\infty, -1) \cup \left(\frac{1}{2}, \infty\right)$$

Find the values of a and b .

Examination continues overleaf...

- (c) The position of jet ski relative to a stationary ship is plotted on the ship's radar. One unit on the radar represents a distance of 1 metre along the sea level, and the origin represents the ship's position.

The jet ski travels at a constant speed and is observed at point A on a position vector of $-20\mathbf{i} + 36\mathbf{j}$. One second later, the jet ski is observed at point B on a position vector of $-2\mathbf{i} + 51\mathbf{j}$.

A crew member on the ship observes the motion of the jet ski using binoculars for ten seconds starting from point A , without taking his eyes off of the jet ski.

- i. Find the position vector of the jet ski after the ten seconds. **2**
 - ii. Hence find the angle through which the crew member's binoculars moves through during this ten second interval, correct to the nearest minute. **2**
- (d)
- i. Express $2\sqrt{3}\cos x - 2\sin x$ in the form $R\cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$ and $R \geq 0$. **2**
 - ii. Hence sketch the graph of **3**

$$y = \frac{1}{2\sqrt{3}\cos x - 2\sin x} \quad \text{for } -\frac{\pi}{6} \leq x \leq \frac{11\pi}{6}$$

showing all important features.

Examination continues overleaf...

Question 13 (14 Marks)

Commence a NEW Writing Booklet

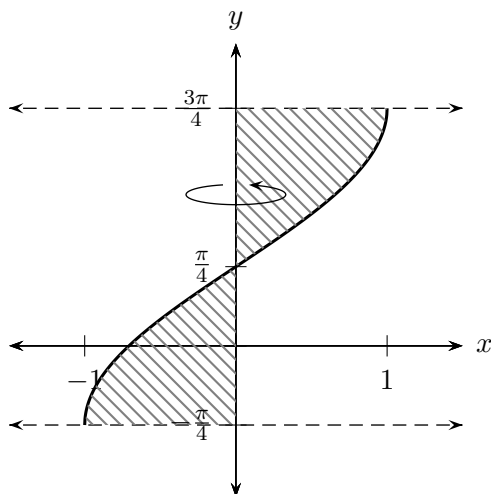
Marks

- (a) A saltwater fish tank contains 250 litres of liquid. To keep the water clean, a pump draws out 15 litres per minute. To keep the amount of liquid in the tank constant, 15 litres of a saline solution with a salinity of 32.4 grams per litre is pumped into the tank every minute. Assume that the salt dissolves evenly into the solution in the tank.



Let $Q(t)$ be the amount of dissolved salt in the fish tank after t minutes. The tank initially contains 8070 grams of salt dissolved in the liquid.

- i. Show that $\frac{dQ}{dt} = \frac{24300 - 3Q}{50}$. **1**
 - ii. Hence solve the differential equation for the solution to the amount of salt $Q(t)$ in the tank at any time. **2**
- (b) The region bounded by the curve $f(x) = \sin^{-1}(x) + \frac{\pi}{4}$ and the y -axis between the lines $y = -\frac{\pi}{4}$ and $y = \frac{3\pi}{4}$, is rotated one complete revolution about the y -axis.



- i. Show that $\sin^2\left(x - \frac{\pi}{4}\right) = \frac{1}{2}(1 - \sin 2x)$. **2**
- ii. Hence, or otherwise, find the volume of the solid of revolution. Give your answer as an exact value. **2**

Examination continues overleaf...

- (c) The spread of a flu through a population is modelled by the equation

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$

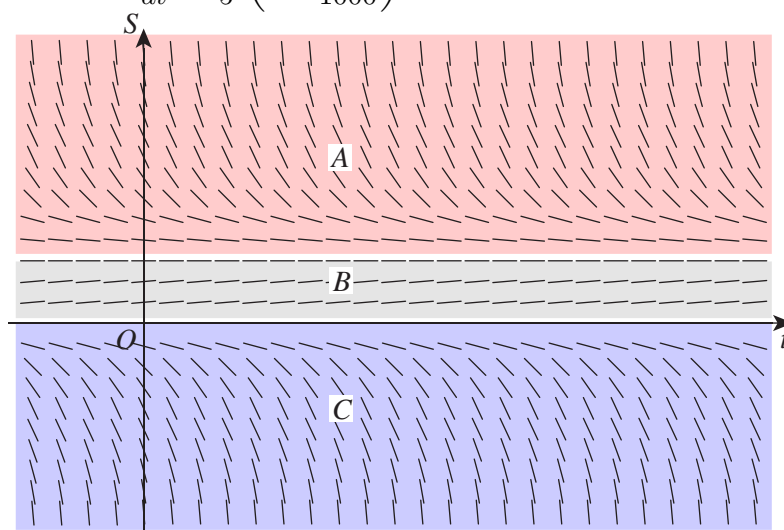
where S is the total number of people infected after t days.

- i. Show that the given equation for S satisfies the differential equation

3

$$\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$$

- ii. The slope field of $\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$ is shown.

2

The three regions A , B and C in which a solution curve can be found, have been labelled and shaded correspondingly.

State which region the given solution curve S can exist in and justify your answer with reference to its constant solutions.

- iii. Using this model, find on which day the rate of increase reaches a maximum. Give your answer correct to the nearest day.

2

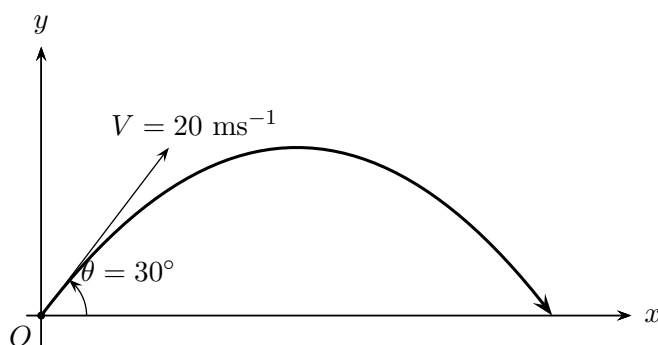
Examination continues overleaf...

Question 14 (14 Marks)

Commence a NEW Writing Booklet

Marks

- (a) Eileen throws a rock from a fixed point O on level ground, with velocity $V = 20 \text{ ms}^{-1}$ at an angle of $\theta = 30^\circ$ with the horizontal.



The flight path of the rock is given by

$$\begin{cases} x_r = Vt \cos \theta \\ y_r = -\frac{1}{2}gt^2 + Vt \sin \theta \end{cases}$$

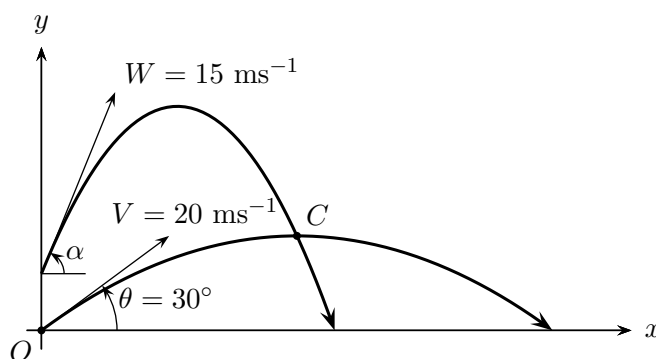
(Do NOT prove this)

where t is the time in seconds after the stone was thrown. (Take $g = 10 \text{ ms}^{-2}$)

- i. Find the maximum height that the rock reaches.

2

Nathan is standing on jungle gym, 3 metres directly above the point O . Nathan throws the stick with velocity $W = 15 \text{ ms}^{-1}$ at angle α above the horizontal, such that it passes through point C where the stone reaches its maximum height.



- ii. Prove that the position of stick is given by

3

$$\underline{r}_s = \begin{pmatrix} 15t \cos \alpha \\ -5t^2 + 15t \sin \alpha + 3 \end{pmatrix}$$

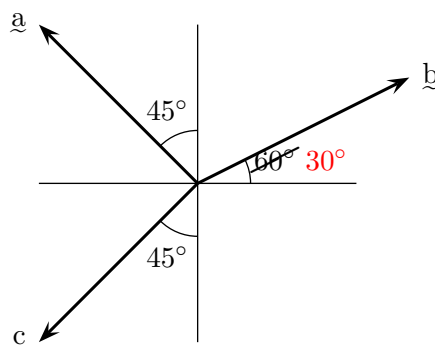
(Take $g = 10 \text{ ms}^{-2}$)

- iii. Determine whether the stick collides with the rock at point C . Justify your answer with calculations.

2

Examination continues overleaf...

- (b) A game of tug-of-war involves three teams who each attach their rope to a weight to pull in their direction as shown. Let $\underline{a}$, $\underline{b}$ and $\underline{c}$ be the pulling force of Team A, Team B and Team C respectively.



Correction: the angle 60° was corrected to 30° .

It is given that $|\underline{c}| = 200$ N. All teams are unable to move the weight from its starting position in any direction.

- i. Show that

2

$$\underline{a} = \left(-\frac{1}{\sqrt{2}} |\underline{a}| \right) \underline{i} + \left(\frac{1}{\sqrt{2}} |\underline{a}| \right) \underline{j}$$

and

~~$$\underline{b} = \left(\frac{1}{2} |\underline{b}| \right) \underline{i} + \left(\frac{\sqrt{3}}{2} |\underline{b}| \right) \underline{j}$$~~

$$\underline{b} = \left(\frac{\sqrt{3}}{2} |\underline{b}| \right) \underline{i} + \left(\frac{1}{2} |\underline{b}| \right) \underline{j}$$

Correction: the component form of $\underline{b}$ using the angle 30° .

- ii. Briefly explain why $\underline{a} + \underline{b} + \underline{c} = \underline{0}$.
- iii. Hence find the exact value of $|\underline{a}|$ and $|\underline{b}|$.

1

4

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

NESA STUDENT #:

Class (please ✓)

☐ 12MXX.1 – Mr Sekaran

☐ 12MAX.2 – Mrs Bhamra

☐ 12MXX.2 – Ms Ham

☐ 12MAX.3 – Mr Lam

☐ 12MAX.1 – Mr Ho

☐ 12MAX.4 – Mr Sun

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)

Suggested Band 6 Responses

1. (A) 2. (A) 3. (C) 4. (D) 5. (D)
6. (B) 7. (B) 8. (C) 9. (A) 10. (B)

Question 11 (Ho)

- (a) i. (1 mark)

✓ [1] for final answer.

$$\begin{aligned}\text{Ways} &= (15 - 1)! \\ &= 14!\end{aligned}$$

- ii. (2 marks)

✓ [1] for correct number of arrangements.

✓ [1] for final answer.

Let E be the event where Aidan, Christopher, Daniel and Matthew are seated together.

$$\begin{aligned}P(E) &= \frac{(12 - 1)! \times 4!}{14!} \\ &= \frac{1}{91} \\ P(\overline{E}) &= 1 - \frac{1}{91} \\ &= \frac{90}{91}\end{aligned}$$

- (b) i. (2 marks)

✓ [1] for correct value of a .

✓ [1] for correct value of c .

$$\begin{aligned}\alpha + \beta + \gamma &= -\frac{12}{a} \\ 6 &= -\frac{12}{a} \\ a &= -2\end{aligned}$$

$$(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = \alpha^2 + \beta^2 + \gamma^2$$

$$\begin{aligned}36 - 2\left(\frac{c}{-2}\right) &= 32 \\ c &= -4\end{aligned}$$

$$\therefore a = -2 \text{ and } c = -4.$$

- ii. (2 marks)

✓ [1] for final answer.

$(x + 2)$ is a factor of $P(x)$:

$$\begin{aligned}P(-2) &= 0 \\ -8(-2) + 48 - 2(-4) + d &= 0 \\ \therefore d &= -72\end{aligned}$$

- (c) (3 marks)

✓ [1] for correct quadratic equation for t .

✓ [1] for one correct pair of solutions for θ .

✓ [1] for all correct solutions for θ .

Using $t = \tan \theta$:

$$\begin{aligned}2\left(\frac{2t}{1+t^2}\right) &= 2 + 4\left(\frac{1-t^2}{1+t^2}\right) \\ 4t &= (2+2t^2) + (4-4t^2) \\ 2t^2 + 4t - 6 &= 0 \\ t^2 + 2t - 3 &= 0 \\ (t+3)(t-1) &= 0\end{aligned}$$

$$t = -3 \text{ or } t = 1$$

$$\tan \theta = -3 \text{ or } \tan \theta = 1$$

$$\begin{aligned}\text{If } \tan \theta = -3 : \quad \text{ref } \angle &= 71^\circ 34' \\ \theta &= 108^\circ 26' \text{ or } 288^\circ 26'\end{aligned}$$

$$\begin{aligned}\text{If } \tan \theta = 1 : \quad \text{ref } \angle &= 45^\circ \\ \theta &= 45^\circ \text{ or } 225^\circ\end{aligned}$$

$$\therefore \theta = 45^\circ, 108^\circ 26', 225^\circ \text{ or } 288^\circ 26'$$

(d) (3 marks)

- ✓ [1] for correct binomial expansion.
- ✓ [1] for correct expressions for c_0 and c_2 .
- ✓ [1] for final result.

$$\begin{aligned}\left(px^3 - \frac{q}{x^2}\right)^9 &= \sum_{r=0}^9 \binom{9}{r} (px^3)^{9-r} (-9x^{-2})^r \\ &= \sum_{r=0}^9 \binom{9}{r} (-1)^r p^{9-r} q^r x^{27-5r}\end{aligned}$$

For the x^{17} term:

$$\begin{aligned}27 - 5r &= 17 \\ r &= 2 \\ c_2 &= \binom{9}{2} p^7 q^2\end{aligned}$$

For the x^{27} term:

$$\begin{aligned}27 - 5r &= 27 \\ r &= 0 \\ c_0 &= \binom{9}{0} p^9\end{aligned}$$

Given $c_0 = c_2$:

$$\begin{aligned}(1)p^9 &= (36)p^7 q^2 \\ p^9 - 36p^2 q^2 &= 0 \\ p^7(p^2 - 36q^2) &= 0 \\ p^7(p - 6q)(p + 6q) &= 0\end{aligned}$$

$$p = 0, p - 6q = 0 \text{ or } p + 6q = 0$$

But given $p > 0$ and $q > 0$.Hence $p - 6q = 0$ only.

(e) i. (1 mark)

✓ [1] for final proof.

$$\begin{aligned}\text{LHS} &= \frac{1}{(n+2)!} - \frac{(n+1)}{(n+3)(n+2)!} \\ &= \frac{(n+3) - (n+1)}{(n+3)(n+2)!} \\ &= \frac{2}{(n+3)!}\end{aligned}$$

 $\therefore \text{LHS} = \text{RHS}$

ii. (3 marks)

- ✓ [1] for proving the base case
- ✓ [1] for applying the inductive step
- ✓ [1] for final proof

Base case: Prove true for $n = 1$.

$$\begin{aligned}\text{LHS} &= \frac{1 \times 2}{(1+2)!} \\ &= \frac{1}{3} \\ \text{RHS} &= 1 - \frac{2^{1+1}}{(1+2)!} \\ &= \frac{1}{3}\end{aligned}$$

 $\therefore \text{LHS} = \text{RHS}$, and the statement is true for $n = 1$.**Inductive step:**Assume true for $n = k$.

$$\begin{aligned}\frac{1 \times 2}{3!} + \frac{2 \times 2^2}{4!} + \frac{3 \times 2^3}{5!} + \dots \\ + \frac{k \times 2^k}{(k+2)!} = 1 - \frac{2^{k+1}}{(k+2)!}\end{aligned}$$

Prove true for $n = k + 1$:**Question 12** (Sun)

$$\begin{aligned} \text{RTP: } \frac{1 \times 2}{3!} + \frac{2 \times 2^2}{4!} + \frac{3 \times 2^3}{5!} + \dots & \quad (\text{a}) \\ + \frac{(k+1) \times 2^{k+1}}{(k+3)!} &= 1 - \frac{2^{k+2}}{(k+3)!} \end{aligned}$$

i. (2 marks)

✓ [1] for correct domain.

✓ [1] for correct range.

$$\begin{aligned} \text{LHS} &= \frac{1 \times 2}{3!} + \frac{2 \times 2^2}{4!} + \frac{3 \times 2^3}{5!} + \dots \\ &+ \frac{k \times 2^k}{(k+2)!} + \frac{(k+1) \times 2^{k+1}}{(k+3)!} \end{aligned}$$

$$\begin{aligned} -1 &\leq \frac{1}{6}x \leq 1 \\ -6 &\leq x \leq 6 \end{aligned}$$

 $\therefore$ Domain is $-6 \leq x \leq 6$

By the inductive step:

$$\begin{aligned} \text{LHS} &= \left(1 - \frac{2^{k+1}}{(k+2)!}\right) + \frac{(k+1) \times 2^{k+1}}{(k+3)!} \\ &= 1 - \frac{(k+3) \times 2^{k+1}}{(k+3)!} + \frac{(k+1) \times 2^{k+1}}{(k+3)!} \\ &= 1 + \frac{[-(k+3) + (k+1)] \times 2^{k+1}}{(k+3)!} \\ &= 1 + \frac{(-2) \times 2^{k+1}}{(k+3)!} \\ &= 1 - \frac{2^{k+2}}{(k+3)!} \end{aligned}$$

$$\begin{aligned} 0 &\leq \cos^{-1} \theta \leq \pi \\ 0 &\leq 4 \cos^{-1} \theta \leq 4\pi \\ -\frac{\pi}{2} &\leq 4 \cos^{-1} \theta - \frac{\pi}{2} \leq \frac{7\pi}{2} \end{aligned}$$

 $\therefore$ Range is $-\frac{\pi}{2} \leq y \leq \frac{7\pi}{2}$

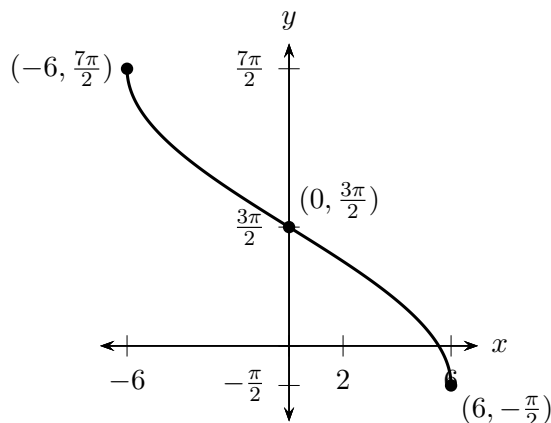
ii. (2 marks)

✓ [1] for shape

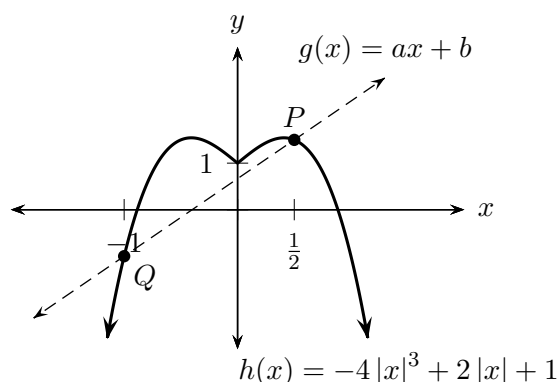
✓ [1] for endpoints and y-intercept

$\therefore$ LHS = RHS, and the statement is true for $n = k + 1$.

Hence by mathematical induction, the statement is true for all positive integers n .



(b) i. (1 mark)

✓ [1] for final sketch of $h(x)$.

ii. (2 marks)

✓ [1] for correct value of a .✓ [1] for correct value of b . P and Q are the points of intersection of $g(x)$ and $h(x)$.

$$h\left(\frac{1}{2}\right) = -4\left|\frac{1}{2}\right|^3 + 2\left|\frac{1}{2}\right| + 1$$

$$= \frac{3}{2}$$

$$\therefore P\left(\frac{1}{2}, \frac{3}{2}\right)$$

$$h(-1) = -4|-1|^3 + 2|-1| + 1$$

$$= -1$$

$$\therefore Q(-1, -1)$$

$$a = \frac{-1 - \frac{3}{2}}{-1 - \frac{1}{2}}$$

$$= \frac{5}{3}$$

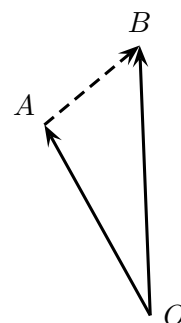
Substitute $Q(-1, -1)$ into $g(x) = \frac{5}{3}x + b$:

$$-1 = \frac{5}{3}(-1) + b$$

$$b = \frac{2}{3}$$

$$\therefore a = \frac{5}{3} \text{ and } b = \frac{2}{3}.$$

(c) i. (2 marks)

✓ [1] for correct vector for $\overrightarrow{AB}$ ✓ [1] for correct position vector $\overrightarrow{OC}$ 

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} -2 \\ 51 \end{pmatrix} - \begin{pmatrix} -20 \\ 36 \end{pmatrix}$$

$$= \begin{pmatrix} 18 \\ 15 \end{pmatrix}$$

Let C be the position of the jet ski after 10 seconds.

$$\overrightarrow{OC} = \overrightarrow{OA} + 10\overrightarrow{AB}$$

$$= \begin{pmatrix} -20 \\ 36 \end{pmatrix} + 10 \begin{pmatrix} 18 \\ 15 \end{pmatrix}$$

$$= \begin{pmatrix} 160 \\ 186 \end{pmatrix}$$

Hence the position vector after 10 seconds is $160\mathbf{i} + 186\mathbf{j}$.

ii. (2 marks)

✓ [1] for substitution into formula

✓ [1] for correct value for θ Let θ be the angle.

$$\overrightarrow{OA} \cdot \overrightarrow{OC} = |\overrightarrow{OA}| |\overrightarrow{OC}| \cos \theta$$

$$\begin{pmatrix} -20 \\ 36 \end{pmatrix} \cdot \begin{pmatrix} 160 \\ 186 \end{pmatrix}$$

$$= (\sqrt{(-20)^2 + 36^2})(\sqrt{160^2 + 186^2}) \cos \theta$$

$$3496 = (\sqrt{1696})(\sqrt{60196}) \cos \theta$$

$$\cos \theta = \frac{3496}{(\sqrt{1696})(\sqrt{60196})}$$

$$\therefore \theta = 69^\circ 45'$$

Hence the angle through which the crew member's binoculars moves is $69^\circ 45'$.

(d) i. (2 marks)

✓ [1] for correct value for R ✓ [1] for correct value for α

$$2\sqrt{3}\cos x - 2\sin x = R\cos(x + \alpha).$$

$$= R(\cos x \cos \alpha - \sin x \sin \alpha)$$

Comparing the coefficients of $\cos x$:

$$2\sqrt{3} = R\cos \alpha \quad (12.1)$$

Comparing the coefficients of $\sin x$:

$$2 = R\sin \alpha \quad (12.2)$$

Equations (12.1)² + (12.2)²:

$$R^2(\sin^2 \alpha + \cos^2 \alpha) = 12 + 4$$

$$R^2 = 16$$

$$\text{since } R \geq 0: \quad R = 4$$

Equations (12.2) $\div$ (12.1):

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

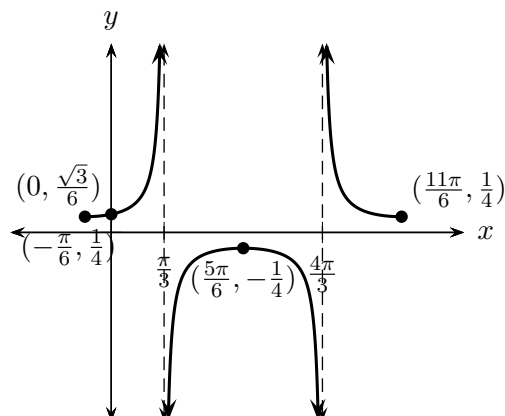
$$\therefore 2\sqrt{3}\cos x - 2\sin x = 4\cos\left(x + \frac{\pi}{6}\right).$$

ii. (3 marks)

✓ [1] for correct asymptotes

✓ [1] for correct turning points

✓ [1] for correct shape

**Question 13** (Lam)

(a) i. (1 mark)

✓ [1] for final proof

Rate of inflow = 486 g/min

Rate of outflow = $\frac{Q}{250} \times 15$ g/min

$$\frac{dQ}{dt} = 486 - \frac{15Q}{250}$$

$$= \frac{24300}{50} - \frac{3Q}{50}$$

$$\therefore \frac{dQ}{dt} = \frac{24300 - 3Q}{50}$$

ii. (2 marks)

✓ [1] for correct integral

✓ [1] for final result

$$\int \frac{1}{24300 - 3Q} dQ = \int \frac{1}{50} dt$$

$$-\frac{1}{3} \ln(24300 - 3Q) + c_1 = \frac{1}{50}t + c_2$$

$$\ln(24300 - 3Q) = -\frac{3}{50}t + c$$

$$24300 - 3Q = e^{-\frac{3}{50}t + c}$$

$$-3Q = Ae^{-\frac{3}{50}t} - 24300$$

$$Q = Be^{-\frac{3}{50}t} + 8100$$

where B is some constantAt $t = 0$, $Q = 8070$:

$$8070 = B + 8100$$

$$B = -30$$

$$\therefore Q(t) = 8100 - 30e^{-\frac{3}{50}t}$$

(b) i. (2 marks)

- ✓ [1] for significant progress
 ✓ [1] for final proof

$$\begin{aligned}\text{LHS} &= \left(\sin x \cos \frac{\pi}{4} - \cos x \sin \frac{\pi}{4} \right)^2 \\ &= \frac{1}{2}(\sin x - \cos x)^2 \\ &= \frac{1}{2}(\sin^2 x + \cos^2 x - 2 \sin x \cos x) \\ &= \frac{1}{2}(1 - \sin 2x) \\ \therefore \text{LHS} &= \text{RHS}\end{aligned}$$

ii. (2 marks)

- ✓ [1] for correct integral
 ✓ [1] for correct exact volume

$$\begin{aligned}y &= \sin^{-1}(x) - \frac{\pi}{4} \\ y + \frac{\pi}{4} &= \sin^{-1}(x) \\ x &= \sin\left(y + \frac{\pi}{4}\right)\end{aligned}$$

Let V be the volume of the solid of revolution about the y -axis:

$$V = \pi \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \left[\sin\left(y - \frac{\pi}{4}\right) \right]^2 dy$$

From part (i):

$$\begin{aligned}&= \pi^2 \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{2}(1 - \sin 2x) dy \\ &= \frac{\pi^2}{2} \left[x + \frac{1}{2} \cos 2x \right]_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \\ &= \frac{\pi^2}{2} \left[\left(\frac{3\pi}{4} + \frac{1}{2} \cos \frac{3\pi}{2} \right) - \left(-\frac{\pi}{4} + \frac{1}{2} \cos \left(-\frac{\pi}{2} \right) \right) \right] \\ &= \frac{\pi^2}{2} \left[\left(\frac{3\pi}{4} + 0 \right) - \left(-\frac{\pi}{4} + 0 \right) \right] \\ \therefore V &= \frac{\pi^2}{2}\end{aligned}$$

(c) i. (3 marks)

- ✓ [1] for correct derivative
 ✓ [1] for significant progress
 ✓ [1] for final proof

$$\begin{aligned}\frac{dS}{dt} &= \frac{-2000(-0.4 \times 199e^{-0.4t})}{(1 + 199e^{-0.4t})^2} \\ &= \frac{2000}{1 + 199e^{-0.4t}} \left(\frac{2}{5} \times \frac{199e^{-0.4t}}{1 + 199e^{-0.4t}} \right) \\ &= \frac{S}{5} \times \frac{2(1 + 199e^{-0.4t} - 1)}{1 + 199e^{-0.4t}} \\ &= \frac{S}{5} \times \frac{2(1 + 199e^{-0.4t}) - 2}{1 + 199e^{-0.4t}} \\ &= \frac{S}{5} \left(2 - \frac{1}{1000} \times \frac{2000}{1 + 199e^{-0.4t}} \right)\end{aligned}$$

$$\therefore \frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$$

ii. (2 marks)

- ✓ [1] for correct constant solutions
 ✓ [1] for correct region

From part (i), when $\frac{dS}{dt} = 0$:

$$\frac{S}{5} \left(2 - \frac{S}{1000} \right) = 0$$

$$S = 0 \text{ or } S = 2000$$

$\therefore$ The logistic curve S is bounded by the two constant solutions $S = 0$ and $S = 2000$.

When $t = 0$:

$$\begin{aligned}S &= \frac{2000}{1 + 199e^0} \\ &= 10\end{aligned}$$

The initial value $S = 10$ is within the interval $[0, 2000]$.

Hence, the solution curve S is in region B .

iii. (2 marks)

- ✓ [1] for correct value of S
- ✓ [1] for final integer solution

The graph of $\frac{dS}{dt}$ is a concave down parabola with roots $S = 0$ and $S = 2000$.

$$\begin{aligned} S_{\text{vertex}} &= \frac{0 + 2000}{2} \\ &= 1000 \end{aligned}$$

When $S = 1000$:

$$1000 = \frac{2000}{1 + 199e^{-0.4t}}$$

$$\begin{aligned} 1 + 199e^{-0.4t} &= 2 \\ e^{-0.4t} &= \frac{1}{199} \end{aligned}$$

$$\begin{aligned} t &= -\frac{5}{2} \ln \left(\frac{1}{199} \right) \\ &= 13.233... \end{aligned}$$

Hence the rate of increase is a maximum on the 14th day.

Question 14 (Bhamra)

(a) i. (2 marks)

- ✓ [1] for correct value of t
- ✓ [1] for correct maximum height

$$\begin{aligned} \dot{y}_r &= -gt + V \sin \theta \\ &= -10t + 20 \sin 30^\circ \\ &= -10t + 10 \end{aligned}$$

Maximum height occurs when $\dot{y}_r = 0$:

$$\begin{aligned} 0 &= -10t + 10 \\ t &= 1 \end{aligned}$$

When $t = 1$:

$$\begin{aligned} y_r &= -\frac{1}{2}(10)(1)^2 + 20(1) \sin 30^\circ \\ &= 5 \end{aligned}$$

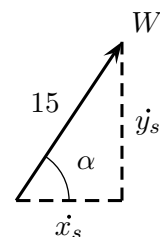
$\therefore$ The maximum height of the rock is 5 metres.

ii. (3 marks)

- ✓ [1] for correct $\underline{\mathbf{a}}_{\mathbf{s}}$
- ✓ [1] for correct $\underline{\mathbf{v}}_{\mathbf{s}}$
- ✓ [1] for final proof

$$\underline{\mathbf{a}}_{\mathbf{s}}(t) = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$$

$$\underline{\mathbf{v}}_{\mathbf{s}}(t) = \begin{pmatrix} c_1 \\ -10t + c_2 \end{pmatrix}$$



When $t = 0$:

$$\dot{x}_s = 15 \cos \alpha$$

$$\dot{y}_s = 15 \sin \alpha$$

$$\therefore \underline{v}_s(t) = \begin{pmatrix} 15 \cos \alpha \\ -10t + 15 \sin \alpha \end{pmatrix}$$

$$\underline{r}_s(t) = \begin{pmatrix} 15t \cos \alpha + c_3 \\ -5t^2 + 15t \sin \alpha + c_4 \end{pmatrix}$$

When $t = 0$, $x = 0$ and $y = 3$.

$$\therefore \underline{r}_s(t) = \begin{pmatrix} 15t \cos \alpha \\ -5t^2 + 15t \sin \alpha + 3 \end{pmatrix}$$

iii. (2 marks)

✓ [1] for appropriate calculations

✓ [1] for correct conclusion

For the rock at point C :

From part (i), $t = 1$.

$$\begin{aligned} x_r &= 20(1) \cos 30^\circ \\ &= 10\sqrt{3} \end{aligned}$$

If the stick collides with the rock at point C :

$$\text{when } t = 1, x_s = 10\sqrt{3}$$

From part (ii), $x = 15t \cos \alpha$:

$$\begin{aligned} 10\sqrt{3} &= 15(1) \cos \alpha \\ \cos \alpha &= \frac{2\sqrt{3}}{3} \end{aligned}$$

But $-1 \leq \cos \alpha \leq 1$

$\therefore$ There is no such value of α for which the the horizontal displacement of the stick is $x_s = 10\sqrt{3}$ when $t = 1$.

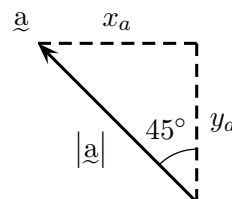
Hence the rock and stick do not collide at C since they reach the horizontal position of C at different times.

(b) i. (2 marks)

✓ [1] for component form of $\underline{a}$

✓ [1] for component form of $\underline{b}$

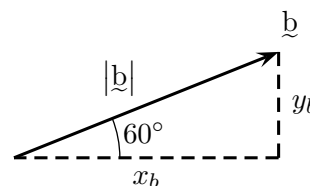
The corrected solutions of Question 14 (b) using the angle 30° is provided in red text within the next section.



$$\begin{aligned} x_a &= -|a| \cos 45^\circ \\ &= -\frac{1}{\sqrt{2}} |a| \end{aligned}$$

$$\begin{aligned} y_a &= \frac{1}{\sqrt{2}} |a| \\ &= \frac{1}{\sqrt{2}} |a| \end{aligned}$$

$$\therefore \underline{a} = \left(-\frac{1}{\sqrt{2}} |a|\right) \underline{i} + \left(\frac{1}{\sqrt{2}} |a|\right) \underline{j}$$



$$\begin{aligned} x_b &= |b| \cos 60^\circ \\ &= \frac{1}{2} |b| \end{aligned}$$

$$\begin{aligned} y_b &= |b| \sin 60^\circ \\ &= \frac{\sqrt{3}}{2} |b| \end{aligned}$$

$$\therefore \underline{b} = \left(\frac{1}{2} |b|\right) \underline{i} + \left(\frac{\sqrt{3}}{2} |b|\right) \underline{j}$$

ii. (1 mark)

✓ [1] for correct explanation

The forces of vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ are in equilibrium as the weight's motion remains constant at rest.

Hence the resultant force is zero:

$$\underline{a} + \underline{b} + \underline{c} = 0$$

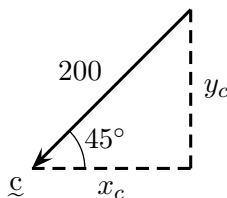
iii. (4 marks)

✓ [1] for component form of $\underline{c}$

✓ [1] for simultaneous equations

✓ [1] for correct values of $|\underline{a}|$

✓ [1] for correct values of $|\underline{b}|$



$$\begin{aligned} x_c &= -200 \cos 45^\circ \\ &= -100\sqrt{2} \end{aligned}$$

$$\begin{aligned} y_c &= -200 \sin 45^\circ \\ &= -100\sqrt{2} \end{aligned}$$

$$\underline{c} = (-100\sqrt{2})\underline{i} + (-100\sqrt{2})\underline{j}$$

From part (ii):

$$\left(-\frac{1}{\sqrt{2}} |\underline{a}| \right) + \left(\frac{\frac{1}{2} |\underline{b}|}{\frac{\sqrt{3}}{2} |\underline{b}|} \right) + \left(\frac{-100\sqrt{2}}{-100\sqrt{2}} \right) = 0$$

$$\therefore \frac{-1}{\sqrt{2}} |\underline{a}| + \frac{1}{2} |\underline{b}| - 100\sqrt{2} = 0 \quad (14.1)$$

$$\frac{1}{\sqrt{2}} |\underline{a}| + \frac{\sqrt{3}}{2} |\underline{b}| - 100\sqrt{2} = 0 \quad (14.2)$$

Equation (14.1) + (14.2):

$$\frac{1}{2} |\underline{b}| (1 + \sqrt{3}) - 200\sqrt{2} = 0$$

$$|\underline{b}| = \frac{400\sqrt{2}}{1 + \sqrt{3}}$$

$$= \frac{400\sqrt{2}(1 - \sqrt{3})}{1^2 - (\sqrt{3})^2}$$

$$= \frac{400(\sqrt{2} - \sqrt{6})}{-2}$$

$$\therefore |\underline{b}| = 200(\sqrt{6} - \sqrt{2}) \quad N \quad (14.3)$$

Substitute equation (14.3) into (14.1):

$$-\frac{1}{\sqrt{2}} |\underline{a}| + \frac{1}{2} \times 200(\sqrt{6} - \sqrt{2}) - 100\sqrt{2} = 0$$

$$\frac{1}{\sqrt{2}} |\underline{a}| = 100(\sqrt{6} - 2\sqrt{2})$$

$$|\underline{a}| = 100(2\sqrt{3} - 4)$$

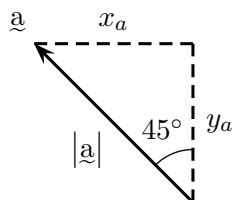
$$\therefore |\underline{a}| = 200(\sqrt{3} - 2) \quad N$$

Corrected solution for Question 14 (b)
(Bhamra)

b i. (2 marks)

✓ [1] for component form of $\underline{a}$

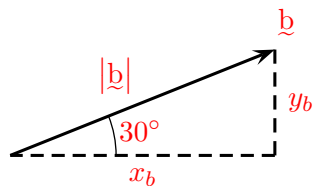
✓ [1] for component form of $\underline{b}$



$$\begin{aligned}x_a &= -|\underline{a}| \cos 45^\circ \\&= -\frac{1}{\sqrt{2}} |\underline{a}|\end{aligned}$$

$$\begin{aligned}y_a &= \frac{1}{\sqrt{2}} |\underline{a}| \\&= \frac{1}{\sqrt{2}} |\underline{a}|\end{aligned}$$

$$\therefore \underline{a} = \left(-\frac{1}{\sqrt{2}} |\underline{a}|\right) \underline{i} + \left(\frac{1}{\sqrt{2}} |\underline{a}|\right) \underline{j}$$



$$\begin{aligned}x_b &= |\underline{b}| \cos 30^\circ \\&= \frac{\sqrt{3}}{2} |\underline{b}|\end{aligned}$$

$$\begin{aligned}y_b &= |\underline{b}| \sin 30^\circ \\&= \frac{1}{2} |\underline{b}|\end{aligned}$$

$$\therefore \underline{b} = \left(\frac{\sqrt{3}}{2} |\underline{b}|\right) \underline{i} + \left(\frac{1}{2} |\underline{b}|\right) \underline{j}$$

ii. (1 mark)

✓ [1] for correct explanation

The forces of vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ are in equilibrium as the weight's motion remains constant at rest.

Hence the resultant force is zero:

$$\underline{a} + \underline{b} + \underline{c} = 0$$

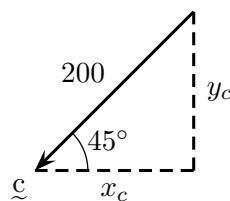
iii. (4 marks)

✓ [1] for component form of $\underline{c}$

✓ [1] for simultaneous equations

✓ [1] for correct values of $|\underline{a}|$

✓ [1] for correct values of $|\underline{b}|$



$$\begin{aligned}x_c &= -200 \cos 45^\circ \\&= -100\sqrt{2}\end{aligned}$$

$$\begin{aligned}y_c &= -200 \sin 45^\circ \\&= -100\sqrt{2}\end{aligned}$$

$$\underline{c} = (-100\sqrt{2}) \underline{i} + (-100\sqrt{2}) \underline{j}$$

From part (ii):

$$\left(-\frac{1}{\sqrt{2}} |\underline{a}|\right) + \left(\frac{\sqrt{3}}{2} |\underline{b}|\right) + \left(-100\sqrt{2}\right) = 0$$

$$\therefore \frac{-1}{\sqrt{2}} |\underline{a}| + \frac{\sqrt{3}}{2} |\underline{b}| - 100\sqrt{2} = 0 \quad (14.4)$$

$$\frac{1}{\sqrt{2}} |\underline{a}| + \frac{1}{2} |\underline{b}| - 100\sqrt{2} = 0 \quad (14.5)$$

Equation (14.4) + (14.5):

$$\frac{1}{2} |\mathfrak{b}| (\sqrt{3} + 1) - 200\sqrt{2} = 0$$

$$|\mathfrak{b}| = \frac{400\sqrt{2}}{\sqrt{3} + 1}$$

$$= \frac{400\sqrt{2}(\sqrt{3} - 1)}{(\sqrt{3})^2 - 1^2}$$

$$= \frac{400(\sqrt{6} - \sqrt{2})}{2}$$

$$\therefore |\mathfrak{b}| = 200(\sqrt{6} - \sqrt{2}) \quad N \quad (14.6)$$

Substitute equation (14.6) into (14.4):

$$\frac{-1}{\sqrt{2}} |\mathfrak{a}| + \frac{\sqrt{3}}{2} \times 200(\sqrt{6} - \sqrt{2}) - 100\sqrt{2} = 0$$

$$\frac{1}{\sqrt{2}} |\mathfrak{a}| = 100(\sqrt{18} - \sqrt{6}) - 100\sqrt{2}$$

$$|\mathfrak{a}| = 100\sqrt{2}(2\sqrt{2} - \sqrt{6})$$

$$\therefore |\mathfrak{a}| = 100(4 - 2\sqrt{3}) \quad N$$